



Oscillatory gamma activity mediates the pathway from socioeconomic status to language acquisition in infancy

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ABSTRACT

Background: The predictable path to child language acquisition is largely constrained by both brain maturation and environmental experience. The synchronized activity of large numbers of neurons gives rise to macroscopic brain oscillations on an electroencephalogram (EEG). It has been found that neural oscillations at rest in the gamma frequency band (25–45 Hz) are associated with development of different cognitive systems, including language. Although the etiology of language is explained by genetically driven brain maturation factors, environment plays a significant role. Specifically, candidate pathways from environment to language development include sociodemographic factors, primarily socioeconomic status (SES) which is likely to exert its effects on language development through other factors, such as parenting style.

Despite these assumptions, no studies have so far examined the interrelation between brain maturation factors such as gamma frequency oscillatory activity, environmental factors such as SES, and language acquisition.

Aim and method: In a longitudinal study of 84 Italian typically developing infants, we measured the power of oscillatory gamma activity as mediator between SES and language acquisition. Baseline EEG and information about SES were collected when infants were aged 6 months. Children were followed-up longitudinally to measure expressive vocabulary and Mean Length of Utterance (MLU) at 24 months.

Results: The mediation model showed that SES is associated with gamma power which, in turn, is associated with expressive language at age 24 months. A higher SES predicted an increase in left central gamma power which, in turn, predicted better language scores.

Conclusions: These results confirm the predictive role of gamma activity oscillatory activity on later language acquisition, suggesting a specific role for these oscillatory mechanisms in language development milestones such as vocabulary development and early word combination. Furthermore, they suggest that SES differences in brain activity may be apparent at early stages of life and affect later language skills. If replicated, our findings could contribute to identifying highest-risk children and may prompt cost-effective preventive/treatment strategies.

Abbreviations: SES, socioeconomic status; MLU, Mean Length of Utterance; PSD, power spectral density; SEM, Structural Equation Modeling; RMSEA, Root Mean Square Error of Approximation; CFI, Comparative Fit Index; SRMR, Standardized Root Mean Square Residual

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1. Introduction

Language is a fundamental and complex human attribute. Surprisingly, acquisition of so complex a system is relatively fast and efficient, with the first two years of life being crucial to this process (Kuhl, 2004). By this age, most children experience an impressive burst in lexical acquisition allowing them to own a large expressive vocabulary (> 50 words; on average 150–250 words (Caselli, Pasqualetti, & Stefanini, 2016; Rescorla, Frigerio, Sali, Spataro, & Longobardi, 2014), and to understand more words than those they can produce. At this age children start combining words, showing initial evidence of productive grammar acquisition. This early predictable path to language acquisition has been shown to be largely constrained by both brain maturation and environmental experience.

A major area of research in basic neuroscience has shown that neurons in the human brain generate complex patterns of oscillatory activity defined as periodic and rhythmic shifting between high and low excitability states (Schroeder, Lakatos, Kajikawa, Partan, & Puce, 2008). At the level of neural ensembles, the synchronized activity of large numbers of neurons gives rise to macroscopic oscillations on an electroencephalogram (EEG). This synchronized oscillatory neuronal activity has been proposed to play a role in the development and maturation of cortical networks (Uhlhaas et al., 2009; Uhlhaas, Roux, Rodriguez, Rotarska-Jagiela, & Singer, 2010). Specifically, high-frequency neuronal oscillations, such as oscillations within the gamma frequency range (> 25 Hz), play a role in establishing precise synchronization of distributed neural responses (Buzsáki & Draguhn, 2004), and gamma power may reflect coordination across widespread cortical regions. Given the neural resources responsible for language are spread throughout the brain, the role of gamma frequency oscillations in coordinating disparate brain regions might be crucial for language acquisition.

Increased power within the gamma frequency range in response to selected stimuli has been linked to different psychological, cognitive and perceptual processes such as object recognition (e.g., Rodriguez et al., 1999), perceptual learning (e.g., Gruber, Müller, & Keil, 2002), attention (e.g., Fan et al., 2007), memory (e.g., Tallon-Baudry, Bertrand, Peronnet, & Pernier, 1998) and language (e.g., Crone et al., 2001; Meyer, 2017; Pulvermüller, Preissl, Lutzenberger, & Birbaumer, 1996). Increased synchronization of neural ensembles in the gamma frequency is crucial for perceptual and higher cognitive processes during early development (e.g., Csibra, 2000), too. For example, selective enhancement of induced gamma oscillations was seen in response to specific native-contrast discrimination at age 6 months (e.g., Ortiz-Mantilla, Hämäläinen, Musacchia, & Benasich, 2013; Peña, Pittaluga, & Mehler, 2010), with a general increase reported from age 6 to 12 months indexing a more efficient processing of native language (Ortiz-Mantilla, Hämäläinen, Realpe-Bonilla, & Benasich, 2016).

The reported relationship between stimulus-induced gamma activity and cognitive/perceptual processes has led to studies addressing whether behavioral individual differences in cognitive and perceptual skills are also related to spontaneous gamma frequency oscillations, as shown by EEG recordings at rest (e.g., Prat, Yamasaki, Kluender, & Stocco, 2016; Tarullo et al., 2017; Tierney, Strait, & Kraus, 2014). Gamma power at rest is seen in the absence of sensory stimulation (as opposed to measuring event-related potentials), and for this reason it is easier to obtain in infants and very young children who struggle to tolerate paradigms requiring active engagement. At this age, however, adult-like resting state recordings characterized by immobility and open eyes cannot be obtained (Stroganova & Orekhova, 2007). Therefore, in the case of infants, EEG recordings under visual attention conditions –for example, when infants are looking to a silent video or to an adult blowing soap bubbles– are generally accepted. Since this relatively standard condition is not equal to the resting state in adults, in the present study we use the term “baseline EEG” to describe such EEG recordings during infancy.

The amount of gamma activity during baseline EEG may be a promising tool for contributing to understanding early neural underpinnings of language development. Methodologically, there is no need for an experimental paradigm to engage infants. In addition, infancy – specifically age 6 months – appears to be the right time during development to assess gamma power during baseline EEG. Specific developmental trajectories of baseline gamma power are documented in the literature showing a progressive increase from age 2 to 6 months (Pivik et al., 2019), a decrease from age 6 to 24 months (Tierney, Gabard-Durnam, Vogel-Farley, Tager-Flusberg, & Nelson, 2012) and an increase again from age 3 to 4 years, with a peak at age 4–5 years (Takano & Ogawa, 1998). Gamma activity in these age ranges has been interpreted as an index of neural maturity and an indicator of neural processes shaping the developing neural architecture in ways that promote efficient cognitive and linguistic functions (e.g., Benasich, Gou, Choudhury, & Harris, 2008; Brito, Fifer, Myers, Elliott, & Noble, 2016; Gou, Choudhury, & Benasich, 2011; Tomalski et al., 2013).

Benasich et al. (2008) analyzed the correlations between frontal gamma power in baseline EEG at age 16, 24 and 36 months and concurrent linguistic and cognitive abilities and showed that, at age 24 months baseline EEG gamma power correlated with both receptive and expressive linguistic skills. This is the age when toddlers are facing a critical window for expressive vocabulary. At age 36 months, when there is less variability in expressive vocabulary, gamma power was associated with receptive skills only. Interestingly, the same measures of baseline EEG frontal gamma power at age 16 to 36 months were additionally found to predict preschool language outcomes (Gou et al., 2011). Results suggested an association with phonological memory (non-word repetition) measured at 4 years and syntactic skills at 5 years. More recently, correlations between parietal gamma power and linguistic abilities were found in newborns (Bruto et al., 2016). Sixty-six full-term infants were tested within 96 h from birth with an EEG recording during active sleep and were re-tested at age 15 months for language and memory skills. A correlation between neonatal gamma power and later language comprehension and working memory emerged.

The predictable path of language acquisition is not only constrained by brain maturation and biological factors but also by environmental experience. Specifically, candidate pathways from environment to language development include sociodemographic factors, primarily socioeconomic status (SES). SES is a well-established predictor of language at many different stages of development and it has been associated with all domains of linguistic competencies and, to a greater extent, with lexical-semantic knowledge and phonological awareness (Noble, McCandliss, & Farah, 2007). SES associations have been found both with expressive and receptive

language. For example, high-SES children had larger expressive vocabularies than mid-SES children. Furthermore, SES accounted for individual differences in performance on a variety of tasks tapping specific neurocognitive systems underlying linguistic abilities (Noble et al., 2007). A more recent study (Riva et al., 2017) documented a direct effect of SES and home literacy environment on expressive language skills in a general population sample aged 6 to 11 years.

A large longitudinal study (Reilly et al., 2007, 2010) assessed the concurrent role of putative environmental risk factors upon language skills in the first 2 years of life. Results showed that both sociodemographic factors, such as SES, and individual/biological factors were associated with language acquisition skills early in life, however only a small amount of variance in language was explained by these environmental factors. The majority of variance that remained unexplained could be attributable to other yet unidentified predictors/mediators (Riva et al., 2017). A further aspect that should be considered is that SES is a complex and distal variable that encompasses several indexes such as occupation, education, income, family stress, home literacy, and nutrition (Riva, Marino, Giorda, Molteni, & Nobile, 2015) and is likely to exert its effects on language development through more proximal familial factors, such as parenting style or mother-child dyadic interactions (Perkins, Finegood, & Swain, 2013).

A few studies have investigated the link between SES and spontaneous brain activity in the gamma frequency range and obtained mixed results. Tomalski et al. (2013) measured gamma power in baseline EEG in two groups of infants aged 6–9 months with low and high SES. Infants of parents with low income and disadvantageous occupation displayed reduced frontal low-gamma (21–30 Hz) power. In addition, the association between family income and frontal gamma power in both low (21–30 Hz) and high (31–45 Hz) gamma ranges was confirmed by regression analyses with SES as a continuous variable. However, in a more recent study including newborns (tested within 96 h from birth), Brito et al. (2016) did not find any correlations between gamma power and family SES, including parental education, household composition and family income. Tarullo et al. (2017) explored for the first time the correlations between gamma power in resting EEG and cognitive outcomes (including both verbal and performance IQ) in disadvantaged children from low- and middle-income countries (rural Pakistan), reporting significant associations even in this condition of environmental risk.

The present study explored the interrelation between brain maturation factors such as gamma frequency oscillatory activity, SES, and language acquisition in a community-based sample of 84 Italian infants. We addressed three major questions: (1) Is gamma power at age 6 months associated with language outcome at age 24 months? (2) Does SES in the first six months of life have an effect on gamma power at 6 months? (3) Does gamma power mediate SES effect on language outcome?

A mediation model using Structural Equation Modeling (SEM) was used to examine whether gamma power mediates the path from SES to expressive language outcome, providing model fit information about the consistency of the mediation hypothesis and evidence for the plausibility of the causality assumption. To our knowledge, this is the first study explicitly addressing the hypothesized direction of association links between SES, oscillatory gamma power at age 6 months, and language outcome at age 24 months using the SEM framework.

2. Materials and methods

2.1. Participants

Eighty-four infants (41 males, 43 females) were included in the study. Infants and their parents were recruited via local advertisement and/or physicians in three different hospitals in (blind). All participants were informed about methodology and duration of the study and parents or legal representative gave their written consent as approved by the local ethics committee.

Inclusion criteria were: (1) having native Italian-speaking parents, (2) gestational age ≥ 36 weeks, (3) birth-weight ≥ 2500 g, (4) APGAR scores at birth at 1' and 5' ≥ 7 and (5) Bayley (Bayley & Stoppa, 2006) cognitive score at age 6 months ≥ 7 (weighted score). Infants with any first-degree relatives having a certified diagnosis of intellectual deficiency, language disorders or autism spectrum disorder were excluded from the study. The study was approved by the Ethical and Scientific Committee of (blind). A description of socio-demographic characteristics of the sample is reported in Table 1.

2.2. EEG data acquisition and processing

Baseline EEG data were acquired at age six months and 15 days (± 2 weeks, $M = 197.1$ days; $SD = 13.5$). During recordings, infants were seated on their mothers' lap in an electrically shielded and quite room. Baseline EEGs were collected following an experimental session featuring a passive oddball paradigm, which was part of a different study (blind). A 4-min block of baseline EEG was collected while a research assistant was blowing bubbles to engage infant's attention.

The EEG was recorded from 60 scalp electrodes using Geodesic Sensor Net (Geodesic EEG System 300, Electrical Geodesics, Inc., Eugene, OR; USA), with impedances below 50 Hz. Vertex was used as an online reference. EEG was sampled at 250 Hz and bandpass-filtered (0.1–100 Hz) online. After recording, EEGs were exported and further processed using lab-internal Matlab (MathWorks, Natick, MA) routines and EEGLAB toolbox (Delorme & Makeig, 2004). In a first step, data was bandpass-filtered between 1 and 47 Hz. Noisy channels were identified using the EEGLAB TrimOutlier plugin (identification criteria: $5 < \text{all channels } SD < 100$) and interpolated with a spherical spline, interpolation of a maximum of 12 out of 60 (20%) channels ($M = 3.30$, $SD = 2.75$) was allowed, otherwise data was discharged from subsequent analysis. All the channels were then re-referenced to common average reference offline. Data was further segmented in 1-sec not overlapping epochs. Automatic artefact detection was performed marking for rejection epochs in which EEG values exceeded $\pm 150 \mu V$ (*find abnormal values* EEGLAB function) and epochs corrupted by linear drift (*find abnormal trends* EEGLAB function; setting parameters: $R = 0.3$, max slope = $150 \mu V$) (Delorme, Sejnowski, & Makeig, 2007).

Table 1
Descriptive statistics related to socio-demographic variables.

	N	Min	Max	Mean (SD)
Gender	84			
% Male	41 (48.8%)			
% Female	43 (51.2%)			
Birth order	83	1	4	1.38 (0.62)
% First	56 (67.5%)			
% Second	22 (26.5%)			
% Third or later	5 (6.0 %)			
SES^a	83	30	90	62.77 (14.00)
% Low/Medium	40 (48.2%)			
% High	43 (51.8%)			
Maternal age at delivery	83	20	48	34.04 (4.93)
Paternal age at delivery	82	26	52	36.29 (5.04)
Maternal education^b	83	20	80	58.07 (17.28)
Paternal education^b	82	20	80	50.37 (17.32)

^a SES was scored according to Hollingshead 90-point scale, whereby a score ranging 10–90 was assigned to each parental job and the higher of two scores was used when both parents were employed (Hollingshead, 1975). Scores ranged between 10, corresponding to unskilled workers; 50, corresponding to sales workers; and 90, corresponding to major professionals.

^b Mother's and father's educational levels were scored on an ad-hoc 90-point ordinal scale based on the Italian school system. Scores ranged between 10, corresponding to the fifth grade of elementary school; 50, equivalent to a high-school diploma; and 90, corresponding to post-doctoral degree.

Additional manual artefact inspection was computed and only artefact-free segments were then entered in further analyses. A minimum of 60 artefact-free segments ($M = 117.21$; $SD = 37.26$) was used for subsequent power analysis (data collected from 11 additional infants were discarded at this point due to lack of artifact-free usable data). Power spectral density (PSD) was estimated by means of Welch method (Welch, 1967) with non-overlapping 0.5-sec windows. PSD values for each channel were calculated and averaged across segments. Following previous literature (Thorpe, Cannon, & Fox, 2016), PSD values were log-transformed using the following formula: $\log \text{PSD} = \log_{10}(1 + \text{PSD})$ which offers the advantage to increase normality while preserving the functional range of zero to infinity for log PSD values. Then, the mean power in the gamma (25–45 Hz) frequency band was calculated. This specific frequency range was selected based on previous infant/toddler baseline EEG literature (Benasich et al., 2008; Brito et al., 2016; Pivik et al., 2019; Tierney et al., 2012; Tomalski et al., 2013; Williams et al., 2012). We decided to encompass what is generally referred to as “low” and “high” gamma (Bruto et al., 2016; Tomalski et al., 2013; Williams et al., 2012), since we did not expect specific effects for the two frequency ranges. We selected six clusters of electrodes in frontal, central, and parietal areas (left and right) (see Fig S1 in the supplementary data) and power values were averaged across electrodes within each cluster. PSD values identified as “extreme values” ($>$ than 3 interquartile ranges from the nearer edge of the boxplot) in SPSS (Version 25.0) were discarded to ensure that correlations were not driven by outliers. This procedure had a minimal effect on the data, since only 5 values out of 504 were discarded (0.99%).

2.3. Socioeconomic status (SES)

Parental socioeconomic status was scored according to Hollingshead 90-point scale (Hollingshead, 1975), whereby a score ranging 10–90 was assigned to each parental job, and the higher of two scores was used when both parents were employed. Scores ranged between 10, corresponding to unskilled workers; 50, corresponding to sales workers; and 90, corresponding to major professionals (descriptive statistics are presented in Table 1). This information referring to the first six months of life of infants was collected when study participants were 6 months of age. This variable was entered into preliminary statistical analyses both as continuous variable (correlational analyses) and as dichotomous variable (low/middle SES score ≤ 60 and high SES score > 60).

2.4. Language assessment

Expressive language at 24 months of age ($M = 24.64$ months; $SD = 0.37$) was rated by the parent-administered Language Development Survey (LDS) that has recently been standardized for the Italian population (Rescorla et al., 2014). The LDS consists of a 310-word vocabulary checklist. Parents are asked to identify each word on the checklist that their children use spontaneously. Expressive vocabulary is thus quantified as the total number of produced words. Parents are also asked whether their children have started combining words into sentences and they need to report five of the child's best sentences. Mean length of utterance (MLU) is quantified in terms of words from the reported sentences. Norms are available for the age range 18 to 35 months (Rescorla & Alley, 2001). Language assessment was available for 62 participants; around 74% of the larger sample (descriptive statistics are presented in Table 2). In order to provide a better description of the sample, both raw scores and percentile scores are reported in the table, although only percentile scores were entered in the analyses.

Table 2
Descriptive statistics of linguistic variables.

	N	Min	Max	Mean (SD)
Vocabulary at 24 months (raw score) ^a	62	5	302	152.08 (82.28)
Vocabulary at 24 months (percentile score)	62	5	90	43.15 (22.55)
MLU at 24 months (raw score) ^b	62	0	7.40	2.99 (1.90)
MLU at 24 months (percentile score)	62	5	85	42.68 (29.87)

MLU = Mean Length of Utterance.

^a Expressive vocabulary quantified as the total number of produced words.

^b MLU quantified in terms of words from the sentences reported by parents.

2.5. Statistical analyses

Descriptive statistics on gamma power and Pearson's bivariate correlations and repeated-measure ANOVAs to examine the relationships between study variables were based on SPSS, Version 25.0.

To investigate the contribution of SES to language in children and the potential role of gamma power as a mediator, we used mediation analysis (Fritz & Mackinnon, 2007), as implemented in the MPLUS software package (Muthén & Muthén, 2014). The mediation model tested the hypothesis that language skills (vocabulary and MLU) would be explained by a sequence of potentially causal effects involving SES and gamma frequency band.

Mediation effects were quantified using the bias-corrected 5000 bootstrap technique to assess non-normality in the product coefficient (Fritz & Mackinnon, 2007). Confidence intervals (95% IC) that do not contain zero indicated significant indirect effects (Mackinnon, Lockwood, & Williams, 2004; Taylor, MacKinnon, & Tein, 2008; Tofghi & MacKinnon, 2011). This method offers the best power, confidence interval placement, and overall control for Type I error (Williams & Mackinnon, 2008). SEM simultaneously models all paths, providing more powerful, accurate and robust estimation of mediation effects (Iacobucci, Saldanha, & Deng, 2007) than more traditional tests based on sequential regressions.

While there is no golden rule for the assessment of model fit, it is advisable to report a variety of indexes (Crowley & Fan, 1997) because different indexes reflect different aspects of model fit. To evaluate the goodness-of-fit of the model, we used the following fit indexes: Chi-Square, Root Mean Square Error of Approximation (RMSEA), Comparative Fit Index (CFI) and Standardized Root Mean Square Residual (SRMR). Chi-square is a test of model misspecification assessing the difference in magnitude between the sample and the model estimated variance/covariance matrices. Its significance is sensitive to large sample sizes and easily produces significant results (Bollen, 1989). In order to address these limitations of the Chi-square statistics, several fit indices have been proposed. The RMSEA – which is one of the most often used indices – is an index that takes into account the complexity of the model (Kline, 2013). Values of RMSEA less than or equal to 0.05 indicate a good fit (Browne & Cudeck, 1993). A common practice is to provide the 90% confidence interval (CI) for the RMSEA, which includes the sampling error associated with the estimated RMSEA. A lower bound of 90% CI less than 0.05 as well as an upper limit less than 0.08, indicate a good fit (Browne & Cudeck, 1993). The SRMR is a standardized version of the RMSEA and represents the average residual value derived from the fit of the model covariance matrix to the sample covariance matrix. SRMR values equal to or less than 0.08 are considered good (Hu & Bentler, 1999). Finally, the CFI is a comparative fit index that measures the improvement of the overall fit of the model by comparing the hypothesized model with a more restricted one, which specifies no relations among variables. Values of CFI greater than or equal to 0.95 indicate a good fit (Whitley, Kite, & Adams, 2013). Study variables included in the mediation analyses showed different rates of missing values: SES 1%, EEG data 0%, language measures 26%. Because data across the observation are missing completely at random (Little's MCAR $\chi^2 = 0.642$; $df = 3$; $p = .887$), missing data were taken into account using Full Information Maximum Likelihood estimation, which is the default estimator, to allow for the use of all available data with the inclusion of participants with missing data. Level of significance was set at $p \leq 0.05$.

3. Results

3.1. Descriptive statistics on gamma power

The average frequency spectrum from 1 to 45 Hz across all participants and electrodes of each cluster is displayed in Fig. 1. Descriptive statistics for log-transformed values in the gamma frequency range (25–45 Hz) are reported in Table 3.

Distribution of gamma power across hemispheres and regions was examined by means of a repeated-measure ANOVA model with Hemisphere (2 levels: Left vs. Right) and Region (3 levels: Frontal vs. Central vs. Parietal) as within-subject factors. Results revealed a main effect of Hemisphere, $F(1,78) = 9.732$, $p = .003$ (greater right than left hemisphere power) and a main effect of Region, $F(2,156) = 75.662$, $p < .001$ (greater power in frontal and parietal than central regions, $p_s < .001$).

3.2. Preliminary analyses

We first examined correlations between SES, language, and gamma frequency band. Topographic plots of correlations are shown in Fig. 2. We found a significant correlation between SES and MLU ($r(60) = .33$, $p = .009$) but not Vocabulary ($r(60) = .20$, $p =$

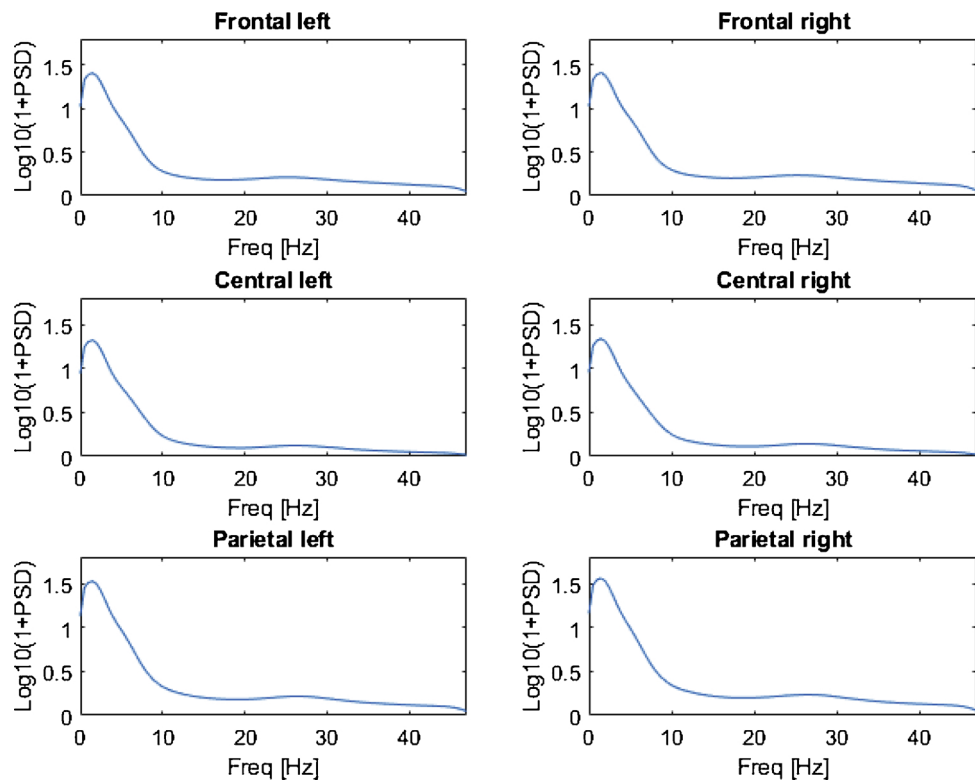


Fig. 1. Log-transformed power spectrum of baseline EEG. Power spectrum was averaged across participants and electrodes of each cluster.

Table 3
Descriptive statistics of power values [after log-transformation] for gamma frequency range in the different clusters of interest.

	Left		Right	
	Mean	SD	Mean	SD
Frontal	.15	.07	.17	.08
Central	.08	.03	.09	.03
Parietal	.15	.05	.17	.08

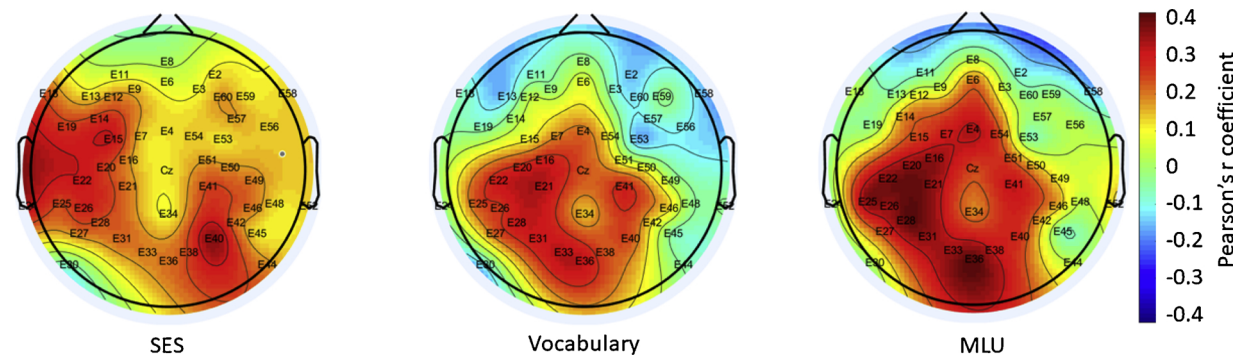


Fig. 2. Topographic plots of correlations. Topographic plots show the relationship between baseline EEG gamma power and SES, Vocabulary and MLU.

.126), with higher SES associated with longer MLU at age 24 months. Significant correlations were also found between SES and power in the gamma frequency range in four out of six channel clusters in the frontal area (frontal left, $r(80) = .23, p = .034$), central area (central left, $r(81) = .30, p = .006$; central right, $r(81) = .23, p = .039$) and parietal area (parietal right, $r(81) = .25, p = .024$): higher familial SES is associated with more positive gamma power. Furthermore, significant associations were found between gamma

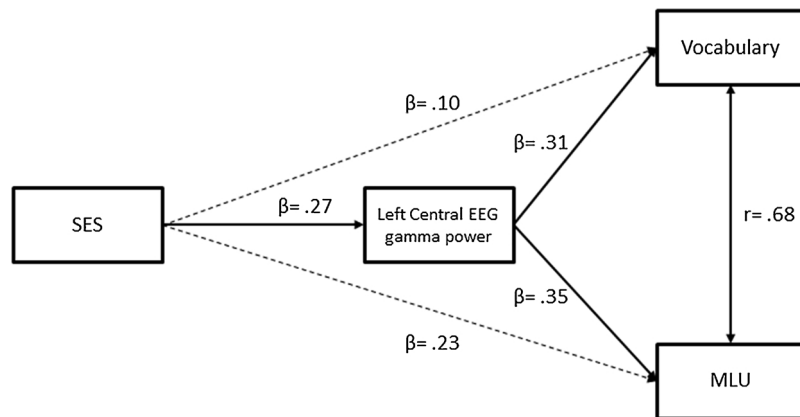


Fig. 3. Mediation model. The model shows Left Central gamma power at age 6 months as a mediator between SES in the first six months of life and toddler language skills at age 24 months.

power on the central area of the left hemisphere and Vocabulary ($r(60) = .32, p = .010$) as well as MLU ($r(60) = .41, p = .001$), with more positive gamma power being correlated with higher expressive vocabulary and longer MLU at age 24 months. Similarly, a significant association was found between gamma power on the parietal area of the left hemisphere and MLU ($r(59) = .26, p = .041$). No other significant correlations emerged between gamma power and language outcome. No significant correlations emerged between EEG measures and other environmental variables (i.e., mother's and father's age at delivery, maternal and paternal education, birth order).

Preliminary analyses were additionally carried out with SES as a dichotomous variable (low/middle SES corresponding to a score ≤ 60 and high SES corresponding to a score > 60). To further explore the relationship between SES and gamma frequency oscillations, we performed a repeated-measures ANOVA with Hemisphere (2 levels: Left vs. Right) and Region (3 levels: Frontal vs. Central vs. Parietal) as within-subject factors and SES (2 levels: low/middle SES vs. high SES) as between-subject factor. The ANOVA revealed a main effect of SES, $F(1,76) = 6.67, p = .012$, confirming that higher family SES is associated with more positive gamma power across regions and hemispheres. In addition, to further explore the relationship between SES and language, we performed a repeated-measures ANOVA with language measure (Vocabulary vs. MLU) as within-subject factor and SES (2 levels: low/middle SES vs. high SES) as between-subject factor. The ANOVA revealed a main effect of SES, $F(1,60) = 4.95, p = .030$ and a significant language measure $\times$ SES interaction, $F(1,60) = 10.81, p = .002$. Separate ANOVAs for each language measure confirmed that MLU scores differed between the two SES groups, $F(1,61) = 8.93, p = .004$, whereas vocabulary scores did not, $F(1,61) = 0.88, p = .351$.

3.3. Testing a mediation model: gamma power as a mediator between SES and toddler language skills

Having shown that individual differences in language outcome are predicted by central gamma power on the left hemisphere, we then used SEM to test the mediation model depicted in Fig. 3, which assumes that SES contributes to central gamma power on the left hemisphere at age 6 months which, in turn, affects language skills at age 24 months.

The mediation model provided good fit to the data [$\chi^2(6) = 59.06, p < .001$; $RMSEA = .000, CI(90\%) = .000-.000$; $CFI = 1.00$; $SRMR = .000$] and explained 19.6% of the variance in language outcome.

Standardized estimates of path coefficients are shown in Fig. 3. The mediation model yielded several significant direct effects. There was a significant path coefficient from central gamma power to Vocabulary scores ($\beta = .31, p = .014$) and MLU ($\beta = .35, p = .001$). Children with more positive central gamma power on the left hemisphere have higher language scores at age 24 months. A significant effect was found from SES to central gamma power ($\beta = .27, p = .007$): higher SES predicts more positive left central gamma power.

More interestingly, using 5000 bootstrap CI 95%, we found a significant indirect effect from SES to MLU score via the left central gamma power ($\beta = .095; SE = .046; 95\% CI = .006-.184; p = .037$): SES was associated with more positive central gamma power at age 6 months which, in turn, influenced syntactic skills at age 24 months.

4. Discussion

The present study investigated the gamma frequency oscillatory activity in a longitudinal community-based sample of 84 Italian infants aged 6 months. Specifically, we were interested in the interdependence of sociodemographic factors such as SES, baseline EEG gamma power at age 6 months – a crucial time in development for brain growth, and later language acquisition at age 24 months – a particularly significant time in linguistic development.

Distribution of gamma power across hemispheres and regions was partially consistent with reports in the literature. We observed greater gamma power in the right vs. left hemisphere, as reported in infants aged 2-6-months (Pivik et al., 2019) and 6 months to 2 years (Tierney et al., 2012). We also observed greater gamma power in the frontal and parietal regions vs. central regions. Higher

levels of gamma power in more anterior regions have been frequently observed in early childhood (e.g., Pivik et al., 2019; Takano & Ogawa, 1998), leading some researchers to focus on these regions (Benasich et al., 2008; Gou et al., 2011; Tierney et al., 2012; Williams et al., 2012). In the present study, we also included central and parietal regions to provide a general overview of the specific regional distribution pattern of gamma activity at age 6 months. Explanations for the resulting pattern (greater power in the right vs. left regions and in the frontal and parietal vs. central regions) might rely on structural changes occurring in the infant brain during the early postnatal period. In particular, since it is well established that gamma oscillations are modulated by GABA-ergic inhibitory interneurons (Başar, 2013; Buzsáki & Wang, 2012), evidence for greater gamma power in the frontal and occipital regions vs. other brain regions is consistent with evidence for a major interneuron migratory pathway that targets the anterior and posterior cortex during late gestation and the first six postnatal months (e.g., Alzu'bi et al., 2017).

Besides these considerations, our first experimental question concerned the association between gamma power in baseline EEG at age 6 months and language outcome (expressive vocabulary and MLU) at 24 months. Overall, we found that infants with greater gamma power at age 6 months displayed a richer expressive vocabulary and produced longer sentences at 24 months. This is consistent with previous literature (Benasich et al., 2008; Brito et al., 2016; Gou et al., 2011) and confirms that gamma oscillatory activity – being involved in neural processes promoting maturation of brain structure and connectivity – appears to be critical for the optimal progress in linguistic development. Despite similarities with previous studies, some specificities of our sample and our results are worthy of note and are further discussed.

First, previous investigations included measures of gamma power collected in children aged 16–36 months (Benasich et al., 2008; Gou et al., 2011) or in newborns aged less than four days (Brito et al., 2016). Based on scientific work on specific early developmental trajectories (Pivik et al., 2019; Takano & Ogawa, 1998; Tierney et al., 2012), we know that baseline EEG gamma power changes dramatically over the first three years of age. Therefore, its associations with later language skills might be expected to change with age, too, suggesting a different role for gamma oscillations on language processes depending on age. Our study is an attempt to fill the current gap in the literature, specifically focusing on baseline EEG gamma power at 6 months, which appears to be a crucial time in development (Pivik et al., 2019; Tierney et al., 2012). Our results confirm that even at age 6 months gamma power is positively correlated with later language development.

Second, the specific topographical pattern of the correlations found in our study might be due to the age of testing or other characteristics of our sample. Brito et al.'s (2016) study found that gamma power in parietal (and not frontal) regions was correlated with later language outcome in newborns. Our results show that at age 6 months correlations with later language outcome seem to involve only central and parietal regions. Studies conducted on toddlers (16–36 months) explored only gamma power in frontal regions and found correlations with concurrent (Benasich et al., 2008) and later (Gou et al., 2011) language skills. In older children or teenagers, correlations with concurrent language or reading skills involved both anterior and posterior regions (Tarullo et al., 2017; Tierney et al., 2014). Taken together, these results suggest a developmental change in the topography of correlations, with associations with language skills being more posteriorly localized in newborns and infants and spreading more anteriorly later on during development. This hypothesis is consistent with some negative results reported in the literature, for example with the reported absence of correlations between frontal gamma power and later language skills in three-month-olds (Levin, Varcin, O'Leary, Tager-Flusberg, & Nelson, 2017). A further aspect to be considered is that, most of the studies reported in the literature focused or at least included children at higher risk for developmental delay, such as children at familial risk for language impairment (Benasich et al., 2008; Gou et al., 2011), for autism spectrum disorder (Levin et al., 2017), socio-culturally disadvantaged children (Tarullo et al., 2017) or children with congenital heart disease (Williams et al., 2012). Together with the research by Brito et al. (2016), our study is the only ones considering a community-based sample without specific risk conditions. The similarity in the topography of correlations between the two studies could be due to this characteristic.

Finally, our study provided evidence for left-lateralized correlations between gamma power at 6 months and language outcome, in line with common knowledge that language is subserved by specialized brain systems localized in the left cerebral hemisphere and with the idea that a disruption of the language lateralization pattern may be implicated in the neurobiology of language impairments (for a review see Bishop, 2013). Oscillations in high frequency bands might be specifically involved in the neural mechanisms providing asymmetric routing between cerebral hemispheres. For example, in a recent study (Cantiani et al., 2019), infants at low risk for language impairment demonstrated a more left-lateralized pattern of gamma oscillatory activity in the auditory cortex during processing of rapidly presented non-speech auditory stimuli, whereas infants at higher risk exhibited a more right-lateralized pattern.

Our second question concerned the association between SES and gamma power in baseline EEG at age 6 months. We found that family SES was positively associated with amount of gamma power, with correlations being significant in different brain areas involving frontal, central and parietal regions. The effect of SES on early gamma power is in line with the Tomalski et al.'s (2013) study, but was not replicated by Brito et al. (2016). The latter interpreted their null finding as depending on the sample composition (general population in which the socioeconomic range, although distributed, did not oversample families in the lowest extremes) or as suggesting that EEG disparities associated with SES-related variables may arise during postnatal experience (since their study included newborns tested within 96 h from birth). Our results allow ruling out the first hypothesis, suggesting that postnatal 6-month exposure to a certain SES is enough to shape the maturation of brain oscillations. The fact that SES distribution in our sample is slightly skewed to the high end (with limited representation of the lowest segment, see Table 1) suggests that it is not only deprivation typical of low SES that affects brain development: SES brain differences are also found across the SES continuum, supporting evidence for a SES gradient association to measures of brain and language.

Our third and final experimental question concerned the role of gamma power in mediating the link between SES and behavioral outcomes in language acquisition during infancy. Our results show a significant indirect effect from SES to MLU score via left central gamma power, indicating that SES was associated with more positive gamma power at age 6 months which, in turn, affected syntactic

skills at 24 months. Although we found a significant correlation between SES and MLU at 24 months, in the mediation model the direct effects between SES and both measures of language outcome at 24 months (expressive vocabulary and MLU) were not significant. This is in line with previous large longitudinal studies (e.g., Reilly et al., 2007) reporting that putative risk factors - including SES - explained a small amount of variation in language at 24 months (overall no more than 7%). Mediators could help explain some of the unexplained variance (Riva et al., 2017). In our case, using gamma power as a mediator of the relationship between SES and MLU allowed to explain about 17% of variance in the development of syntactic skills, a greater proportion of variance compared to previous studies assessing only the effect of the environment risk factors upon language skills (Reilly et al., 2007, 2010). Overall, the environment reflected in SES seems to exert – very early in life – an influence on brain network related to language development.

The study presents some limitations. First, language outcomes were assessed solely by parental report. Although direct measures would be preferable, at this age assessment is highly dependent on the state of the toddler, whereas parents' reports are generally accurate. Parents' reports of expressive vocabulary in toddlers are highly correlated with concurrent and later measures of language development (Rescorla & Alley, 2001) and measures of MLU accurately predict grammar skills, specifically if used before age 3 years (Devescovi & Pizzuto, 1995). Second, it should be pointed out that the results of mediation analyses based on cross-sectional data may be unlikely to accurately reflect longitudinal mediation effects (Cole & Maxwell, 2003). The model tested here appears to be partially cross-sectional, with SES information and baseline EEG being collected at the same time-point (6 months of age). However, SES refers to the first 6 months of life and it could be argued a temporal precedence and ordering of SES and baseline EEG measures. Future longitudinal designs and cross-lagged panel design between measurement times should be considered for replication of the present work. An additional and final limitation concerns SES, which is a broad, distal and often poorly defined variable lacking evidence for the underlying mechanisms that could mediate the relationship between SES and brain functioning. More proximal factors, such as parenting style and early language exposure, might have contributed independently to poorer language capacities and/or to the dysfunction of brain networks (Ellwood-Lowe, Sacchet, & Gotlib, 2016). Even if SES is one of the most replicated environmental risk factors for language and cognitive development, this limitation is common to almost EEG studies on socioeconomic disparities recently reviewed by Pietto, Kamienskowski, and Lipina (2017)), and reflects the need of further studies providing more defined measures of the proximal factors that might disentangle the specific mechanisms responsible for the associations between SES, gamma power, and language development.

In conclusion, the present study has the potential of filling some gaps in the current literature about the developmental trajectories of neural oscillations in baseline EEG in the gamma frequency band. Specifically, we described the topographical distribution of such oscillations at 6 months of age and explored their functional role in the development of language skills at 24 months. The interdependence of gamma frequency oscillatory activity, environmental factors such as SES, and language acquisition are explored for the first time, highlighting a crucial role for brain oscillatory activity as a mediator between SES and language skills. Since a measure of the amount of gamma activity during baseline EEG is relatively easy to obtain in infants (a few minutes of EEG recording are required, in which the infant does not need to be engaged in any experimental paradigm), it can be considered a promising tool for understanding how the infant brain is shaped by environment early in life and how this produces cascading effects on later development. Further larger longitudinal studies are needed to delineate more clearly the early developmental trajectories of gamma oscillatory activity in baseline EEG, to explore developmental changes in the amount of such oscillations and how these changes are associated with later development in populations at high- and low-risk for neurodevelopmental disorders. Our results show that the effect of environment can be observed very early in development, with direct effects on the maturation of the neural networks as reflected in gamma oscillatory activity. Overall, this study and its future directions may open up new perspectives for earlier detection of at-risk infants that would lead to earlier intervention and possibly improve outcomes.

Declaration of Competing Interest

None.

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